Finite-Element Method (FEM)

Boundary value problem: find $\underset{\sim}{u}\ \forall \underset{\sim}{x}$ such that balance of linear momentum

$$\operatorname{div} \underset{\sim}{\sigma} + \rho \overline{\underset{\sim}{b}} = \underset{\sim}{0} \qquad \forall\ \underset{\sim}{x} \in \mathcal{B}$$

(body forces)

is fulfilled under the boundary conditions

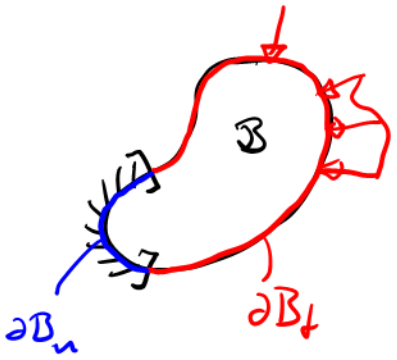
$$\underset{\sim}{u} = \overline{\underset{\sim}{u}} \qquad \text{on } \partial\mathcal{B}_u$$

$$\underset{\sim}{t} = \underset{\sim}{\sigma}\,\underset{\sim}{n} = \overline{\underset{\sim}{t}} \qquad \text{on } \partial\mathcal{B}_t$$

(surface tractions)

with $\partial\mathcal{B}_u \cup \partial\mathcal{B}_t = \partial\mathcal{B}$

$\partial\mathcal{B}_u \cap \partial\mathcal{B}_t = \{\}$



Weak form

Galerkin method, partial integration, Gauss integral theorem, Cauchy theorem yields after some algebraic manipulations

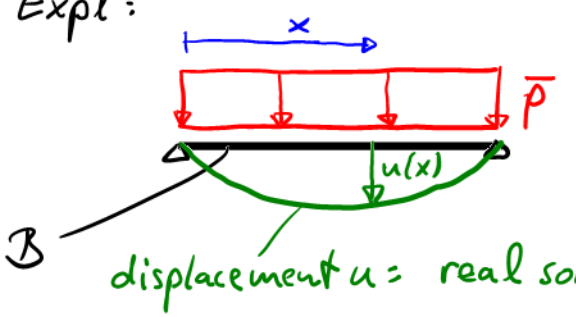
$$G := \int_{\mathcal{B}} \delta\underset{\sim}{\varepsilon} : \underset{\sim}{\sigma}\, dv - \int_{\mathcal{B}} \delta\underset{\sim}{u} \cdot \rho\overline{\underset{\sim}{b}}\, dv - \int_{\partial\mathcal{B}_t} \delta\underset{\sim}{u} \cdot \overline{\underset{\sim}{t}}\, da = 0$$

G^{int} $\qquad$ G^{ext}

with test functions $\delta\underset{\sim}{u}$ a. $\delta\underset{\sim}{\varepsilon} = \frac{1}{2}\left(\operatorname{grad}^T \delta\underset{\sim}{u} + \operatorname{grad}\delta\underset{\sim}{u}\right)$

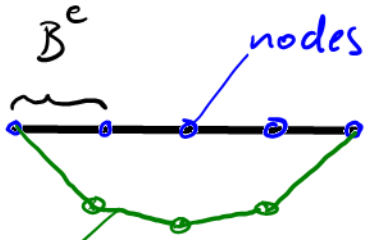
Basic idea of FEM

Expl:



displacement u: real solution

FE-Discretization



u^h: approximated solution field

Direct calculation

- Ansatz: Polynom of 4th order

  $u^h = ax^4 + bx^3 + cx^2 + dx + e$

  quantities to be computed
- Boundary conditions

  $u(0) = u''(0) = u(\ell) = u''(\ell) = 0$

  $p(x) = \bar{p}$

☹ Type of ansatz is not known for arbitrary problems

FE-Method:

- Ansatz: $u^h(x)$ inside $\mathcal{B}^e$ is calculated from interpolation between nodal displacements $\underline{\underline{D}}$

  quantities to be computed
- Boundary conditions

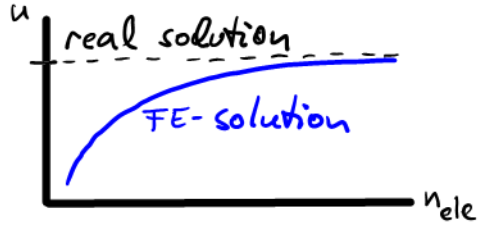
  $u(0) = u(\ell) = 0$

  $p(x) = \bar{p}$

✓ Applicable for arbitrary problems

⇩

FE-convergence



⇐

Basic idea:

$\underset{\sim}{u} \Leftarrow \underset{\sim}{u}^h(\underline{\underline{D}})$ in $\mathcal{B}^{eh}$

$\Rightarrow \mathcal{B} \approx \mathcal{B}^h = \bigcup_{e=1}^{n_{ele}} \mathcal{B}^{eh}$

Linear FEM

Approximation

Displacements: $\underline{\underline{u}} \approx \underline{\underline{N}}_e \underline{\underline{d}}_e =: \underline{\underline{u}}^h$

($\underline{\underline{d}}_e$: nodal displacements; $\underline{\underline{N}}_e$: contains shape functions)

Strains: $\underline{\underline{\varepsilon}} \approx \underline{\underline{B}}_e \underline{\underline{d}}_e =: \underline{\underline{\varepsilon}}^h$

($\underline{\underline{B}}_e$: contains derivatives of shape functions w.r.t. $\underline{\underline{x}}$)

analog for test functions $\delta\underline{u}$ and $\delta\underline{\underline{\varepsilon}}$

Linear-elastic material law

$$\underline{\underline{\sigma}} = \underline{\underline{\mathbb{C}}} : \underline{\underline{\varepsilon}} \rightsquigarrow \underline{\underline{\sigma}} \approx \underline{\underline{\mathbb{C}}}\, \underline{\underline{\varepsilon}}^h = \underline{\underline{\mathbb{C}}}\, \underline{\underline{B}}_e \underline{\underline{d}}_e$$

Discrete form and system of equations

Insert approximations and material law in weak form:

$$G^h = \sum_e G_e^h = 0 \Longrightarrow \sum_e \left\{ \delta \underline{\underline{d}}_e^T \left[\underline{\underline{k}}_e \underline{\underline{d}}_e - \underline{\underline{p}}_e \right] \right\} = 0$$

mit $\underline{\underline{k}}_e = \int_{\mathcal{B}_e} \underline{\underline{B}}_e^T \underline{\underline{\mathbb{C}}}\, \underline{\underline{B}}_e \, dV$ ($\underline{\underline{\mathbb{C}}}$: from material law)

$$\underline{\underline{p}}_e = \int_{\mathcal{B}_e} \underline{\underline{N}}_e^T \underline{\bar{b}} \, dV + \int_{\partial\mathcal{B}_t} \underline{\underline{N}}_e^T \underline{\bar{t}} \, da$$

Representation of sum as matrix multiplication

$$\underline{\underline{D}} = \bigcup_{e=1}^{n_{ele}} \underline{\underline{d}}_e \Rightarrow \delta\underline{\underline{D}}^T \left(\underline{\underline{K}}\,\underline{\underline{D}} - \underline{\underline{P}} \right) = 0 \qquad (1)$$

$\underline{\underline{D}}$: Desired column matrix of nodal displacements

$\underline{\underline{K}}\,\underline{\underline{D}} =: \underline{\underline{F}}^{int}$ (internal forces)

$\underline{\underline{P}} =: \underline{\underline{F}}^{ext}$ (external forces)

with $\underline{\underline{K}} = \bigcup_{e=1}^{n_{ele}} \underline{\underline{k}}_e$

and $\underline{\underline{P}} = \bigcup_e \underline{\underline{p}}_e$ Assembling operator

(1) must be fulfilled for arbitrary values of $\delta\underline{\underline{D}}$

$\Rightarrow$ Global system of equations

$$\underline{\underline{R}} := \underline{\underline{F}}^{int} - \underline{\underline{F}}^{ext} = \underline{\underline{0}} \quad \rightsquigarrow \quad \underline{\underline{K}}\,\underline{\underline{D}} = \underline{\underline{P}}$$

$\rightsquigarrow$ solution yields desired displacements $\underline{\underline{D}}$

Special case linear elasticity:

- $\underline{\underline{R}} = \underline{\underline{0}}$ is linear equation
- $\underline{\underline{K}}$ is independent from $\underline{\underline{D}} \rightsquigarrow$ LINEAR system of eq.

Material model enters

- stiffness matrix $\underline{\underline{\mathbb{C}}} \rightarrow \underline{\underline{K}}$
- Calculation of stresses from $\underline{\underline{\sigma}} = \underline{\underline{\mathbb{C}}}\,\underline{\underline{B}}_e\,\underline{\underline{D}}$

Extension to nonlinear material models

$\underline{\underline{R}} = \underline{\underline{0}}$ becomes nonlinear equation ↯

Newton method

Example problem: $f = f(x)$

- Taylor-expansion

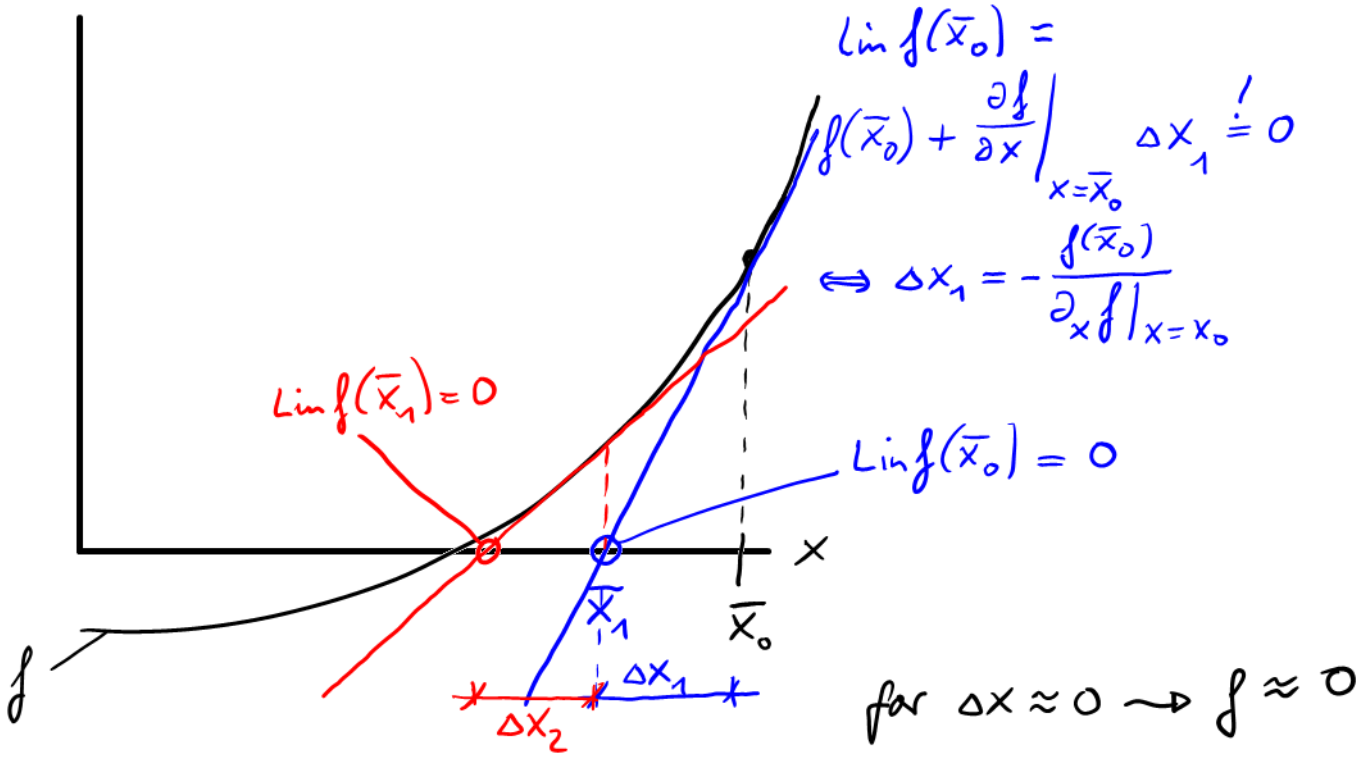
$$f(x+\Delta x) = f(x) + \frac{\partial f}{\partial x}\Delta x + \underbrace{O(\Delta x^2)}_{\text{negligible for small } \Delta x}$$

- Linearized equation $\left(O(\Delta x^2) = 0\right)$:

$$\text{Lin}\, f = f(x) + \underbrace{\frac{\partial f}{\partial x}\Delta x}_{=:\Delta f} = 0$$

Δ-Operator

$$f(g(x)) \rightsquigarrow \Delta f = \frac{\partial f}{\partial g}\underbrace{\Delta g}_{=\frac{\partial g}{\partial x}\Delta x}$$



- Iterative procedure

  Loop over $k = 0, 1, \ldots$ until $|\Delta x| < tol$

  1) $f\big|_{x_k} + \left.\frac{\partial f}{\partial x}\right|_{x_k} \Delta x_{k+1} = 0$ (solve linearized equation)

  2) $x_{k+1} = x_k + \Delta x_{k+1}$ (update)

  3) Wenn $|\Delta x_{k+1}| < tol \rightsquigarrow$ exit (stopping criterion)

  - Initial value shall not be too far from solution
  - Quadratic convergence very close to solution
  - Disadvantage: derivative $\frac{\partial f}{\partial x}$ is required

Application to nonlinear mechanics problem

$$\underline{\underline{R}}(\underline{\underline{D}}) = \underline{\underline{0}}$$

$$\rightsquigarrow \text{Lin}\ \underline{\underline{R}} = \underline{\underline{R}}\Big|_{\underline{\underline{D}}_k} + \left.\frac{\partial \underline{\underline{R}}}{\partial \underline{\underline{D}}}\right|_{\underline{\underline{D}}_k} \Delta \underline{\underline{D}}_{k+1} \overset{!}{=} \underline{\underline{0}} \qquad (1)$$

$$\underbrace{= \underline{\underline{F}}^{int} - \underline{\underline{F}}^{ext}}_{\underline{\underline{R}}} \qquad \underbrace{= \frac{\partial \underline{\underline{F}}^{int}}{\partial \underline{\underline{D}}}}_{\left.\frac{\partial \underline{\underline{R}}}{\partial \underline{\underline{D}}}\right|} = ?$$

Linear increment of internal forces

Inner part of weak form

$G := G^{int} - G^{ext} = 0$ with

$$G^{int} = \int_{\mathcal{B}} \delta \underline{\underline{E}} \cdot \underline{\underline{S}} \, dV \quad \text{mit} \quad \delta \underline{\underline{E}} = \frac{1}{2}\left(\mathrm{Grad}\,\delta \underline{u} + \mathrm{Grad}^T \delta \underline{u}\right)$$

$$\underline{\underline{S}} := \underline{\underline{S}}\left(\underline{\underline{E}}(\underline{u})\right)$$

- Linear increment:

$$\Delta G^{int} = \int_{\mathcal{B}} \underbrace{\left(\Delta \delta \underline{\underline{E}} \cdot \underline{\underline{S}}\right)}_{=0} + \delta \underline{\underline{E}} \cdot \Delta \underline{\underline{S}} \, dV$$

$$= \int_{\mathcal{B}} \delta \underline{\underline{E}} \cdot \underbrace{\left(\frac{\partial \underline{\underline{S}}}{\partial \underline{\underline{E}}} : \Delta \underline{\underline{E}}\right)}_{=: \underline{\underline{\mathbb{C}}}} dV = \Delta G$$

(für kons. äußere Kräfte)

- Matrix notation: $\delta \underline{\underline{E}} \to \delta \underline{\underline{E}}$, $\Delta \underline{\underline{E}} \to \Delta \underline{\underline{E}}$, $\underline{\underline{\mathbb{C}}} \to \underline{\underline{\mathbb{C}}}$

$$\rightsquigarrow \quad \Delta G = \int_{\mathcal{B}} \delta \underline{\underline{E}}^T \, \underline{\underline{\mathbb{C}}} \, \Delta \underline{\underline{E}} \, dV$$

- FE-Diskretization: $\delta \underline{\underline{E}} = \underline{\underline{B}}_e \, \delta \underline{\underline{d}}_e$, $\Delta \underline{\underline{E}} = \underline{\underline{B}}_e \, \Delta \underline{\underline{d}}_e$

$$\rightsquigarrow \quad \Delta G = \sum_e \int_{\mathcal{B}^e} \left(\underline{\underline{B}}_e \, \delta \underline{\underline{d}}_e\right)^T \underline{\underline{\mathbb{C}}} \, \underline{\underline{B}}_e \, \Delta \underline{\underline{d}}_e \, dV$$

$$\Leftrightarrow \quad \Delta G = \sum_e \delta \underline{d}_e^T \underbrace{\int_{B^e} \underline{B}_e^T \underline{\underline{C}} \underline{B}_e \, dV}_{\underline{\underline{k}}^e} \Delta \underline{d}_e$$

- Assembling: $\quad \Delta \underline{D} = \bigcup_e \Delta \underline{d} \quad , \quad \underline{\underline{K}} = \bigwedge_e \underline{\underline{k}}_e$

+ incorporation of Dirichlet boundary conditions yields

$$(1): \; \text{Lin}\, \underline{R} = \underline{R}\Big|_{\underline{D}_k} + \underline{\underline{K}}\Big|_{\underline{D}_k} \Delta \underline{D}_{k+1} \stackrel{!}{=} \underline{0}$$

$$\frac{\partial \underline{R}}{\partial \underline{D}} = \frac{\partial \underline{F}^{int}}{\partial \underline{D}}$$

$$\rightsquigarrow \quad \underline{\underline{K}} \, \Delta \underline{D} = - \underline{R}$$

attention: $\underline{R} \neq \underline{F}^{ext}$

$$\underline{R} = \underline{F}^{int} - \underline{F}^{ext} \; \checkmark$$

"Residual": $\underline{R} = \underline{R}^{int} - \underline{R}^{ext}$ with $\underline{R}^{int} = \bigwedge_e \underline{r}_e^{int}$, $\underline{r}_e^{int} = \int_{B^e} \underline{B}^T \underline{\sigma} \, dV$

Required from material model:

$\underline{\sigma}$ for $\underline{R}$ and $\underline{\underline{C}} = \frac{\partial \underline{\sigma}}{\partial \underline{\varepsilon}}$ for $\underline{\underline{K}}$

Algorithm I:

1. Initialize $k=0$, $\underline{\underline{D}}_{k=0} = \underline{\underline{\tilde{D}}}_0 = ?$

2. Loop over k

   2.1 Loop over e

      2.1.1 $\underline{\underline{k}}^e\big|_{\underline{\underline{D}}_k}$, $\underline{r}^e\big|_{\underline{\underline{D}}_k}$

      2.1.2 sort in $\underline{\underline{K}}$, $\underline{\underline{R}}$

   2.2 Solve $\underline{\underline{K}}\big|_{\underline{\underline{D}}_k} \Delta\underline{\underline{D}}_{k+1} = -\underline{\underline{R}}\big|_{\underline{\underline{D}}_k}$

   2.3 Update $\underline{\underline{D}}_{k+1} = \underline{\underline{D}}_k + \Delta\underline{\underline{D}}_{k+1}$

   2.4 Stop ?

      If $|\Delta\underline{\underline{D}}_{k+1}| \leq tol$, then exit

- Initial value ~~close to~~ solution: decomposition of load in n_{load} load steps $\rightsquigarrow$ $\bar{\underline{f}}_i$, $\bar{\underline{b}}_i$, $\bar{\underline{d}}_i$ with $i = 1 \dots n_{load}$ and set initial value of Newton iteration equal to solution of last load step.
- Initial value for first load step: $\underline{\underline{D}}_{k=0}^{i=0} = \underline{\underline{0}}$

Algorithm II:

1. Initialize $i = 0$, $\underline{\underline{D}}^{i=0} = \underline{\underline{0}}$

2. Loop over $i = 0, 1, \ldots n_{load} - 1$

   2.1 $\underline{\underline{D}}^{i+1}_{k=0} = \underline{\underline{D}}^{i}$

   2.2 Update boundary conditions $\bar{t}_{i+1}$, $\bar{b}_{i+1}$, $\bar{d}_{i+1}$

   2.3 Solve equilibrium using Algorithm I
   $\rightsquigarrow \underline{\underline{D}}^{i+1}$